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FEMO FAQs

The first prompt of the job interview was “tell me about your experience with data management and analysis, especially any programming skills.” The second question was “do you know how to operate chainsaws or have any active red-card qualifications?” I am not a C faller or a computer science major, but between my student seasons on a wildland fire module and some coding for my undergraduate thesis, I managed to land the job with Yosemite’s fire effects crew.

I loved being a fire effects monitor (“FEMO”) precisely because of the many unique skillsets we drew upon and developed to get the job done. The work was physically and mentally challenging, but the sheer variety made it well worth it. One typical week in August might start off with a few grueling days chasing lightning starts with the park’s wildland fire module (“WFM”), transition to some office time coding up a new data management workflow, and end by installing monitoring plots in Yosemite Valley to prepare for a fall prescribed burn. Over the course of my two seasons with Yosemite, I supported on-park fire management efforts as a firefighter, helped steward the park’s long-term fire effects monitoring system, and went on off-park fire assignments with the Yosemite WFM and the US Forest Service Fire Behavior Assessment Team.

It was a great job, but one thing I noticed throughout my work was persistent puzzlement amongst primary fire folks, other agency personnel, and the public about what a “fire effects monitor” does and why our work is important. This confusion is not helped by the fact that historically, “FEMO” is both a red card qualification that can be held by any wildland firefighter and also a distinct federal job series, “Biological Science Technician (Fire Effects Monitor)” with different educational, pay, and duty requirements compared to primary wildland firefighters.* Collectively, these different kinds of FEMOs occupy the gray area between primary and secondary fire, which is both our greatest strength and also sometimes a source of confusion.

I hope to use this essay as an opportunity to answer some FAQs about FEMO work, and in the process, make the case for why you – whether you are a hotshot, an incident commander, an ecologist, or simply a concerned member of the public – should care about, work with, or consider becoming a FEMO yourself!

What is the purpose of a FEMO?

The overarching goal of any FEMO is to collect field data to inform fire management and fire science. The job is not simply to understand the basics of fire behavior and fire effects, but to record observations (rate of spread, percent canopy torch, 1000-hr fuel consumption, etc.) for a given fire in a systematic, reproducible way that allows scientists and land managers to compare

many fires across time, space, and condition. In this sense, FEMOs serve as a key bridge between fire ecology – the academic study of how fires shape ecosystems – and the messy, practical business of fire management.

What brought you to FEMO work?

I caught the fire bug during a stint on AmeriCorps St. Louis's Emergency Response Team, where I had the opportunity to work on prescribed burns with the Missouri Department of Conservation's State Fire Ecologist, Susan Farrington. As both a deeply knowledgeable ecologist and a bad ass burn boss, Susan inspired me to think about fire in a new way and made sure I got my hands sooty while doing it. After AmeriCorps, I went back to school and worked student seasons on a wildland fire module (Apache Kid WFM, Cibola NF), a type of handcrew that specializes in managing remote wilderness fires for ecological resource benefit. Once I graduated with a bachelors in ecology, I qualified for the biological science technician job series and joined the fire effects crew in Yosemite. Inspired by questions that first crystallized for me during my monitoring work, I am now going back to school to study fire's impact on soil microbiology. During my studies, I plan to continue working on fires as a FEMO, both with my local prescribed burn association and as an on-call member of the US Forest Service Fire Behavior Assessment Team (see below).

What background skills do I need to become a FEMO?

Working at the intersection of primary firefighting and ecological monitoring, FEMOs come to the job from a variety of different backgrounds. Some begin in primary fire and start out more comfortable handling a chainsaw or a pulaski than a forestry basal area prism. Others move into FEMO work from more traditional restoration or monitoring crews and must learn the basics of firefighting on the fly. Regardless of your background, almost everyone has something new to learn, and one of the best parts of being a FEMO is the opportunity to work on teams that draw on a variety of expertise and backgrounds, from botany to coding to sawing, to accomplish our objectives.

What is the difference between "Fire Effects Monitor" as a red card qualification and as a job title?

These different qualifications/positions, both bearing the title of "fire effects monitor," reflect different emphases on the relative importance of fireline experience and ecological skills to the monitoring task at hand.

Data collection on an actual fire requires the FEMO to be a trained firefighter well versed in standard fire operations to safely accomplish common incident tasks, from recording rate of spread measurements during firing operations to jumping in to help chase spots. As such, "FEMO" is a red-card qualification requiring completion of a taskbook, and certain "primary" fire crews that specialize in fires managed for resource benefit, such as Type I and Type II wildland fire modules, are required to have a FEMO or two on their roster. On a typical managed fire with my WFM, one day I might be assigned to help our lead FEMO establish photo plots tracking how the fire was impacting a habitat of concern in our burn area; the next shift I could

be back with my saw partner helping to prep line for another burn operation. At the end of the assignment, we would compile our daily observations and plot data into a summary report for the incident management team and district office.

But what happens after your 14 days are up and the crew gets reassigned to the next fire? Who is going to track how the flame lengths you recorded translate into ecosystem impacts months to years down the line? This is when the other kind of a “fire effects monitor” – biological science technicians organized into small five-person “fire effects” crews – swing into action. These crews are much less numerous than WFMs and have been traditionally limited to the Park Service due to the agency’s particular emphasis on long-term ecological monitoring, though other departments may have individuals or crews that perform similar duties. The principal objective of these crews is to revisit burn scars at regular intervals to perform species inventories, track the build-up of fuel, and assess tree mortality. This data is then recorded in a unified national fire effects database used by researchers and managers to guide long-term decision making about fire policy. As such, fire effects crews often emphasize botany, tree physiology, and data skills in addition to fire experience, and the positions used to require a college degree for all crewmembers.* Working on the Yosemite fire effects crew, I helped remeasure decades-old monitoring plots in some of the first federal prescribed burn units in the West. Over the years, data from these plots helped establish the importance of prescribed burning as a management tool in these ecosystems, an important lesson dramatically underscored when the burn units saved the town of Wawona and the Mariposa Grove of Giant Sequoias from the devastating 2022 Washburn fire (Hankin et al., 2023).

Wait, both kinds of FEMOs sound cool! Can I do both?

Just as FEMOs in general straddle the fire operations and fire ecology worlds, so too can the different kinds of FEMO work blend together in practice. Most members of a dedicated Park Service fire effects crew will earn their FEMO red card qualification within a season or two, if they do not have it already. And similarly, some wildland fire modules will draw on their local fire effects crew into order to beef up their monitoring capacity if required for a specific prescribed burn or managed fire.

That said, different fire effects crews and wildland fire modules will lean into different aspects of FEMO work depending on the needs of their home unit. Some WFMs focus heavily on suppression and ignition operations, using their FEMOs primarily to ensure they are following the burn prescription during an operation. Some fire effects crews will spend most of their season doing plot work and rarely go on fire assignments. I was very fortunate to begin my training on a WFM with experienced FEMOs invested in doing more than just taking weather observations, then continue onto a fire effects crew that was particularly well-integrated into a fire management organization that gave us ample opportunities to go out on fires.

The best example of a crew that integrates every aspect of FEMO work at a high level is the US Forest Service’s Fire Behavior Assessment Team (FBAT). The module is basically the wildland fire equivalent of storm-chasers. Its main scientific mission is to collect fire behavior and effects data on extreme fire events that, unlike prescribe burns or fires managed for resource benefit, are usually not accessible for detailed monitoring by academics, fire effects crews, or even normal

WFMs (Vaillant et al., 2014). A typical FBAT roster includes both extremely experienced operational leaders (at least one division-qualified personnel) and scientific experts (fire ecologists and physicists), rounded out by FEMOs and FFT1s to help with data collection. The crew takes detailed fuels, botanical, and forestry measurements usually associated with fire effects crew-style FEMO work but does so on the dynamic operational tempo of an active incident. When I worked with the FBAT on the Dillon Fire last season, we hopped in front of a burnout operation to set up heat-resistant thermal cameras to measure energy transfer dynamics, surveyed impacted endangered Humboldt marten habitat to inform future burning collaborations with the Karuk Tribe, and provided daily observation reports to the incident command team's fire behavior analyst (FBAN) when relevant.

What is the toughest part about being a FEMO?

I will focus here on what it is like to work on a fire effects crew off-incident, since the FEMO work done on active fires by wildland fire modules and other primary firefighters brings many of the same challenges and risks associated with the handcrew work already described by other Backbone contributors.

First, it should be noted that even if you are not on an active fire incident, plot monitoring work can also be physically demanding. Hiking in miles to count seedlings underneath head-high whitethorn brush with no intact canopy cover at the height of summer is not a task for the light-hearted, and the same physical vulnerabilities that plague primary firefighters also to some extent impact FEMOs. Working through my last season with a bone bruise on my knee from a training injury was difficult, and the experience underscored to me that despite no longer being in "primary fire," my livelihood was still heavily dependent on remaining fit and uninjured.

Second, historically* FEMOs on fire effects crews were biological science technicians and therefore not eligible for pay bonuses or any of the other incentives developed to improve firefighter retention. This stung when we were on the line or otherwise contributing to fire response capacity like helping staff engines on our weekends. While I loved that I could work on a science-oriented team while remaining a dialed squad ready to roll out to a lightning bust at a moment's notice, it was frustrating to do the same work on a fire as my co-workers on the engine or handcrew for lower pay.

Why Should I Care About FEMOs?

If you are burn boss, our observations might be the legal document proving you acted within the scope of your prescription.

If you are an academic ecologist, we might collect the raw data you need for your models and studies.

If you are a hotshot, we might be the ones helping to calibrate your spot weather forecasts or hold for your firing operation.

If you are an engine captain, we might be the first line of militia showing up to help on a gnarly initial attack.

If you are a district ranger, you might rely on us to deliver accurate, comprehensive descriptions of the burns and managed fires you authorized on your land.

If you are a member of the public, you need firefighter-scientists like us choking on smoke or crawling under *Ceanothus* brush to collect the data that will allow all of us to make better, more informed decisions about how we live with fire.

*Those planning to apply to FEMO positions in the future should note that starting in 2026, the National Park Service has changed the job's position description and folded fire effects crewmembers into the primary fire "GW" series instead of the old biological science technician series.

References

Hankin LE, Anderson CT, Dickman GJ, Bevington P, Stephens SL. 2023. How forest management changed the course of the Washburn fire and the fate of Yosemite's giant sequoias (*Sequoiadendron giganteum*). *Fire Ecology* 19:40.

Vaillant NM, Ewell CM, Fites-Kaufman JA. 2014. Capturing crown fire behavior on wildland fires—The Fire Behavior Assessment Team in action. *Fire Management Today*:41.